

Simplicity as a Feature: Robust Transition-State Search by Gentlest Ascent Dynamics with Cheap Analytic Hessians

June 25, 2026

Abstract

Machine-learned interatomic potentials (MLIPs) make analytic Hessians cheap (≈ 0.06 s/structure with HIP [9]), so the bottleneck in transition-state (TS) search is no longer the Hessian but the *step*. We use the minimal choice — first-order Gentlest Ascent Dynamics (GAD) on the lowest eigenvector of the Eckart-projected analytic Hessian, a Cartesian step with no trust region, quasi-Newton update, RFO, or eigenvector prediction. Our primary metric is the IRC-validated *intended-reaction success rate* (forward and backward IRC must graph-match the intended reactant and product), on 287 Transition1x test reactions from perturbed-saddle guesses (10–200 pm).

Highlights

- **Crossover, not a point.** GAD is statistically indistinguishable from the strongest Sella baseline when the guess is good (McNemar $p \approx 1.0$, ≤ 50 pm) and pulls significantly ahead as it degrades ($p < 10^{-4}$ by 150–200 pm; **+21.6 pp intended at 200 pm**, 45.3% vs. 23.7%). The same crossover holds for saddle-convergence and across $F_{\max}$ thresholds.
- **Mechanism (corrected).** No method converges to an unintended (wrong-basin) saddle — unintended $\approx 0/287$ for all three. GAD’s edge is recovering the full two-sided reaction more often; Sella’s high-noise loss is non-convergence + one-sided (partial) saddles, not wrong basins.
- **Honest cost.** First-order GAD cannot tighten $F_{\max}$ below ≈ 0.01 eV/Å; no step budget removes it. This plateau lives on the *secondary* (convergence) axis only.
- **Hybrid.** A GAD→Newton switch removes the plateau, wins wall-time per converged TS (1.4–3 $\times$) and geometry (tightest p95 RMSD), and inherits GAD’s chemistry.
- **Scope boundary.** Robustness is demonstrated for *perturbed-saddle* guesses. From a reactant start the GAD family is weak (hybrid $\sim 2\%$) and Sella wins — GAD needs to start near the saddle for the reaction mode to be identifiable.

1 Introduction

- TSs are index-1 saddle points; locating them needs curvature (the Hessian, or at least its leftmost eigenpair). *Ab initio* Hessians are expensive, so optimizers avoid them: dimer [2], eigenvector-following / P-RFO [3, 4, 5], Sella’s iterative partial-Hessian diagonalization [6, 7], and recent ML prediction of the Hessian [10, 9] or its leftmost eigenvector [11].
- **The premise has changed.** With MLIPs the analytic Hessian is cheap (≈ 0.06 s for HIP [9]). The bottleneck becomes the **step**. Trust-region/Newton jumps ($\mathcal{O}(0.1$ Å)) are optimal near a quadratic saddle but fragile from a poor guess.

- **This work.** Given a cheap exact Hessian, discard P-RFO, trust radii, QN, and eigenvector prediction; step with first-order GAD [1]. Score success by IRC-validated *intended* recovery [10, 11], not mere convergence. Thesis: *in the MLIP regime you need not avoid the Hessian — you need to step more carefully with it.*

2 Methods

GAD dynamics. With forces $\mathbf{F} = -\nabla E$ and $\mathbf{v}_1$ the lowest eigenvector of the mass-weighted, Eckart-projected Hessian,

$$\dot{\mathbf{x}} = (\mathbf{I} - 2\mathbf{v}_1\mathbf{v}_1^\top)\mathbf{F} = \mathbf{F} - 2(\mathbf{F}\cdot\mathbf{v}_1)\mathbf{v}_1, \quad \mathbf{x}_{k+1} = \mathbf{x}_k + dt\dot{\mathbf{x}}. \quad (1)$$

- **Step:** fixed- dt explicit Euler ($dt \in \{0.003, 0.005, 0.007\}$; representative 0.005), Cartesian, no history/trust-region/line-search; mode tracked by max overlap; per-atom cap 0.35 Å.
- **Potential:** HIP [9], an $SE(3)$ -equivariant MLIP returning the analytic Hessian (≈ 0.06 s); not trained on Transition1x (held out); Hessian recomputed every step.
- **Eckart projection:** remove the 6 translation/rotation modes in mass-weighted coordinates, diagonalize the reduced $(3N-6)$ Hessian; single most important robustness ingredient. (Cartesian vs. MW step: ≤ 0.35 pp.)
- **Hybrid GAD–Newton:** walk with Eq. (1); near a clean index-1 saddle ($n_{\text{neg}}=1$, small force) switch to a trust-radius-capped eigenvector-following Newton step on the same analytic Hessian.
- **Benchmark:** 287 Transition1x [8] test reactions; start = true saddle + isotropic Gaussian noise $\sigma \in \{10, 30, 50, 100, 150, 200\}$ pm; baseline = Sella [7] (strongest config: Cartesian + Eckart, exact Hessian/step; Appendix D).

Two metrics, kept deliberately separate.

- **Intended-reaction success rate (IRC) — primary.** Did we find the *right* TS? After optimization, follow Sella’s IRC [7] forward/backward and graph-match each endpoint (Open Babel connectivity + VF2) to the labelled reactant/product. Outcomes (after [10, 11]): *intended* (both endpoints match), *partially intended* (one), *unintended* (valid saddle, other reaction), *no reaction* (collapsed to a minimum), *TS error* (no convergence). This is the chemically meaningful score and our headline metric.
- **Saddle-convergence rate — secondary.** Did the optimizer reach *a* first-order saddle at all: $n_{\text{neg}}=1 \wedge F_{\text{max}} < \tau$ (default $\tau = 0.01$ eV/Å; threshold family in Appendix C). A pure-geometry check, necessary but *insufficient* for the intended reaction; reported separately in Section 3.3, not alongside the primary metric.

3 Results

3.1 Primary result: intended-reaction success (IRC) crosses over with guess quality

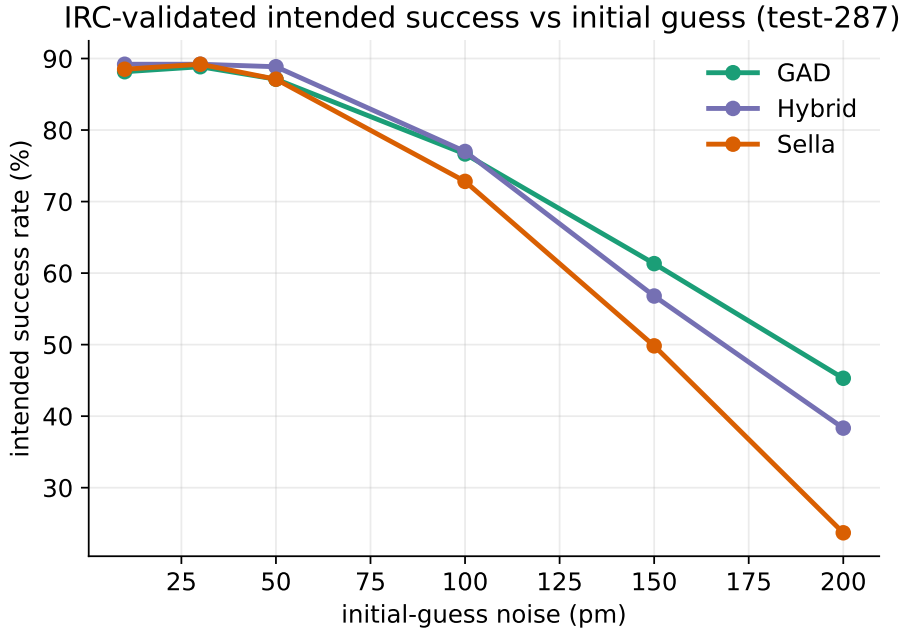


Figure 1: Primary metric: intended-reaction success rate (IRC) vs. initial-guess noise, $n = 287$. *x-axis*: isotropic Gaussian perturbation of the true saddle (10–200 pm). *y-axis*: % of the 287 reactions whose forward and backward IRC graph-match *both* the intended reactant and product. One line each for GAD, the hybrid, and the strongest Sella baseline. The three methods coincide while the guess is good; GAD and the hybrid retain far more intended reactions as it degrades. (Saddle-convergence is shown separately in Fig. 7.)

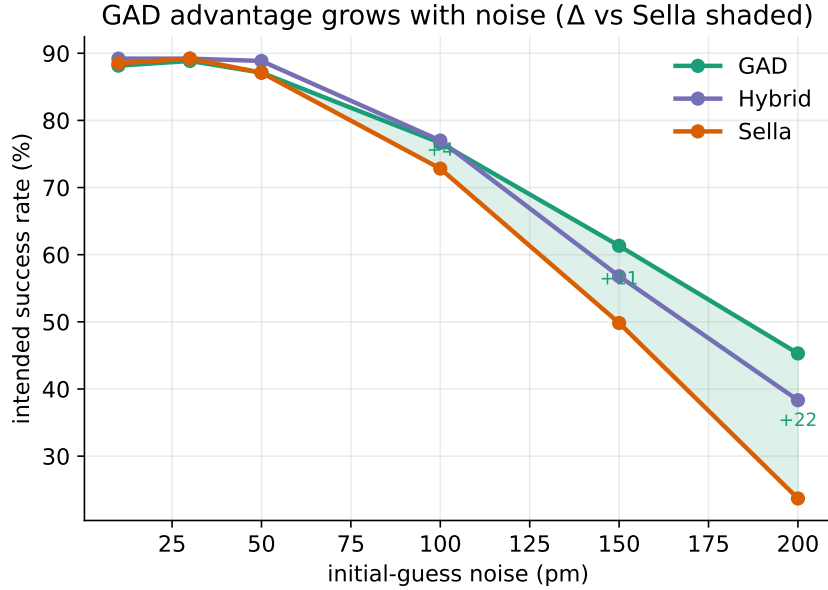


Figure 2: Intended success for all three methods; the GAD–Sella gap is shaded. Paired-McNemar (GAD vs. Sella): indistinguishable ≤ 50 pm ($p=1.0$), emerging at 100 pm ($p=0.10$), highly significant at 150 pm ($p=3.8 \times 10^{-5}$) and 200 pm ($p=1.7 \times 10^{-13}$).

- **Crossover.** Tie at low noise (intended 88–89% at 10 pm); GAD pulls ahead with degradation: $\Delta_{\text{intended}} = +3.9, +11.5, +21.6$ pp at 100/150/200 pm (Fig. 2, Table 1).
- **Significance.** McNemar on paired per-sample intended outcomes: n.s. at ≤ 50 pm ($p=1.0$), emerging at 100 pm ($p=0.10$), highly significant at 150 pm ($p=3.8 \times 10^{-5}$) and 200 pm ($p=1.7 \times 10^{-13}$). The same crossover appears on saddle-convergence and across $F_{\max}$ thresholds (Section 3.3).

Table 1: Primary metric — intended-reaction success rate (IRC), %, $n = 287$, from the fresh per-sample re-validation (GAD $dt=0.003$); Sella = strongest baseline. Saddle-convergence is reported separately in Table 2 (Section 3.3).

Intended success %	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD (ours)	88.2	88.9	87.1	76.7	61.3	45.3
Sella (baseline)	88.5	89.2	87.1	72.8	49.8	23.7
Δ (pp)	−0.3	−0.3	0.0	+3.9	+11.5	+21.6

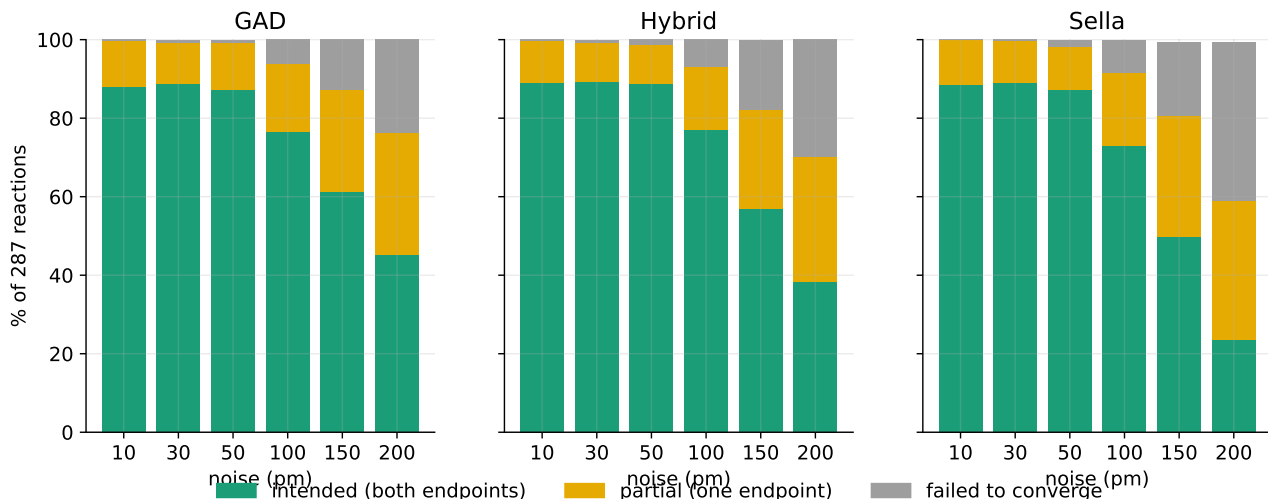


Figure 3: Full IRC outcome breakdown. *Unintended* ≈ 0 for every method at every noise, so the third band is purely non-convergence (TS error / no reaction). Sella both fails to converge *and* lands one-sided (partial) saddles more often at high noise.

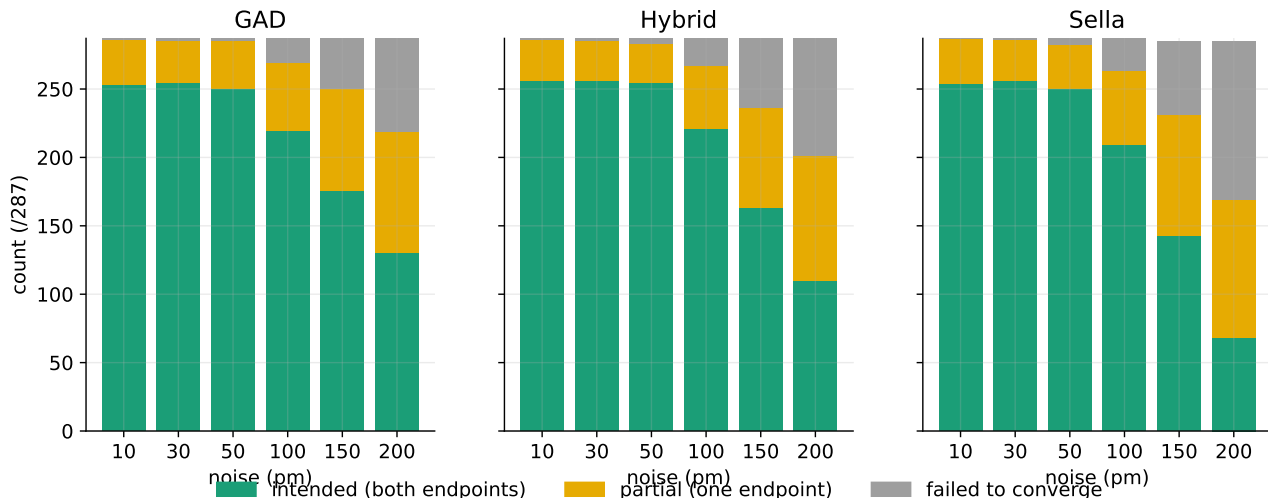


Figure 4: Same breakdown, absolute counts (/287). At 200 pm: intended/partial/failed = GAD 130/89/68 vs. Sella 68/101/116 — Sella’s loss is non-convergence (116 vs. 68) plus one-sided partial saddles (101 vs. 89).

3.2 Mechanism: it is not wrong basins — it is one-sided saddles and non-convergence

- **Unintended** ≈ 0 . The per-sample 5-tier IRC split gives unintended ≈ 0 for every method at every noise: 0 for GAD and the hybrid, and only 2 (150 pm) + 2 (200 pm) for Sella — **4 of 5166** samples total. *We therefore drop the earlier “Newton jumps across basin boundaries to unintended saddles” claim — it is unsupported by the data.*
- **What actually differs** (200 pm illustration; whole curves in Fig. 3): intended/partial/failed = **GAD 130/89/68 vs. Sella 68/101/116**. Sella both fails to converge more *and* lands one-sided (partial) saddles more.

- **Why.** Small in-basin Euler steps preserve the link to *both* reactant and product, so even non-converged GAD geometries IRC-validate on both sides — intended can *exceed* convergence (150 pm: 61.3 vs. 55.1%; 200 pm: 45.3 vs. 40.8%). Larger Newton steps more often drift off one channel ($\rightarrow$ partial) or fail to tighten ($\rightarrow$ failed).
- **Convergence $\neq$ chemistry, stated correctly** (Fig. 5): it is *not* that convergence hides wrong saddles. (i) GAD recovers intended even when not strictly converged; (ii) Sella d=1 vs. d=3 — d=3 *converges more* but is *intended less* at every noise (a stale Hessian yields more partial/failed, not unintended).
- **Non-converged GAD breakdown** (why it fails to *converge*, not why it finds a wrong saddle): at 200 pm, of 163 non-converged, 71 (44%) drift to $n_{\text{neg}} \geq 2$ and 92 (56%) are plateau-orbits at $n_{\text{neg}}=1$ just above threshold (Section 3.3).

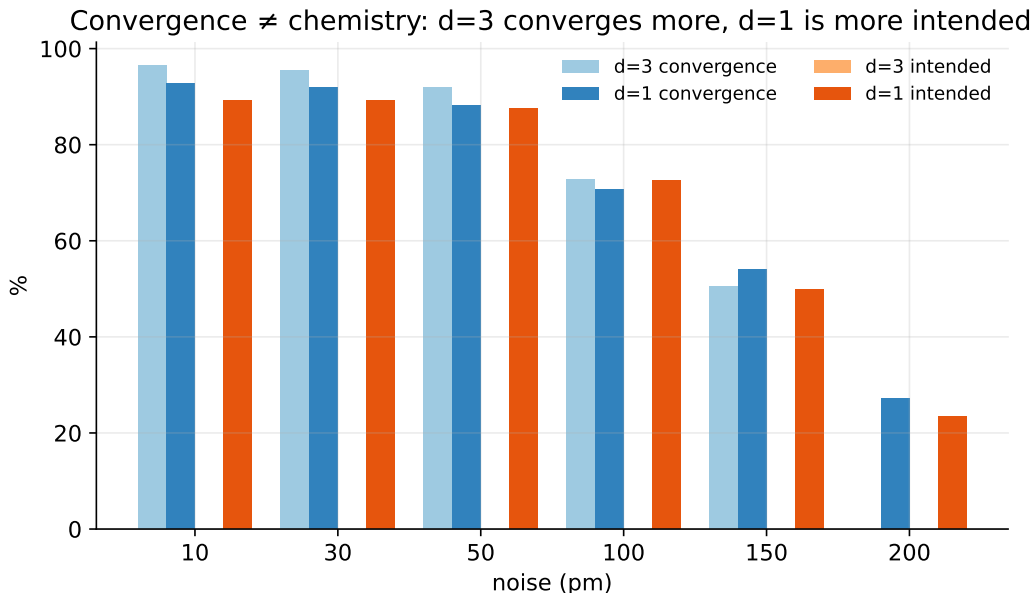


Figure 5: Convergence $\neq$ chemistry. Sella Hess.Freq.=3 converges more than Hess.Freq.=1 at low noise (a) yet recovers fewer intended reactions at every noise (b): the extra “convergences” are one-sided (partial) saddles, not unintended ones.

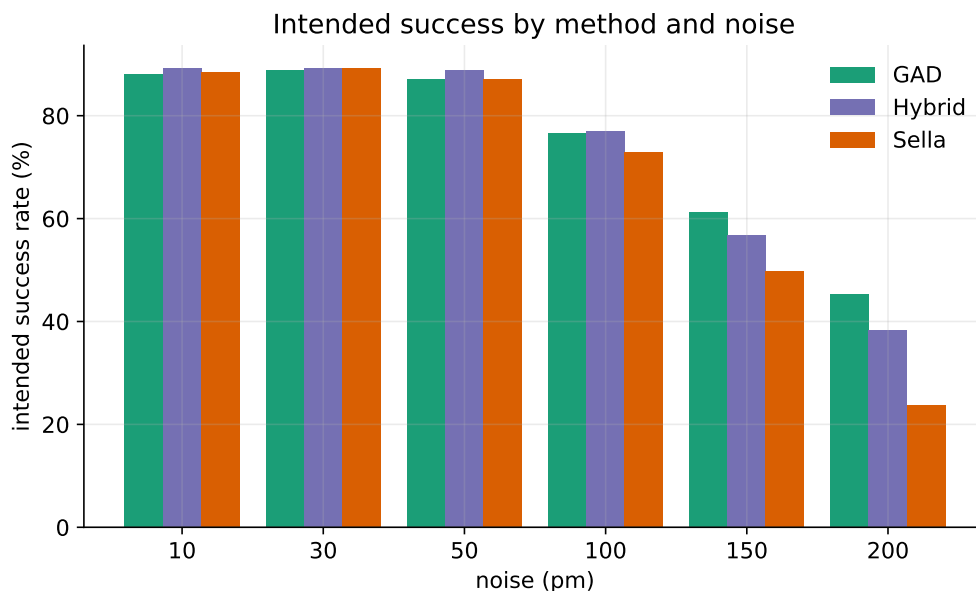


Figure 6: Intended reactions recovered (counts/287). The GAD family overtakes the baseline from 100 pm onward.

3.3 Secondary: saddle-convergence and the $F_{\max}$ plateau

Saddle-convergence ($n_{\text{neg}}=1 \wedge F_{\max}<\tau$) asks only whether the optimizer reached *a* first-order saddle, independent of which reaction it connects.

- **Crossover here too** (Fig. 7, Table 2). At low noise Sella’s Newton step crushes the force fastest, so it leads saddle-convergence (92.7 vs. 89.2% @ 10 pm); GAD overtakes by 150 pm (55.1 vs. 54.0%) and 200 pm (40.8 vs. 27.2%).
- **But convergence $\neq$ chemistry.** Sella’s low-noise convergence lead does *not* convert into intended reactions (Table 1); the d=1/d=3 case makes this explicit (Fig. 5).

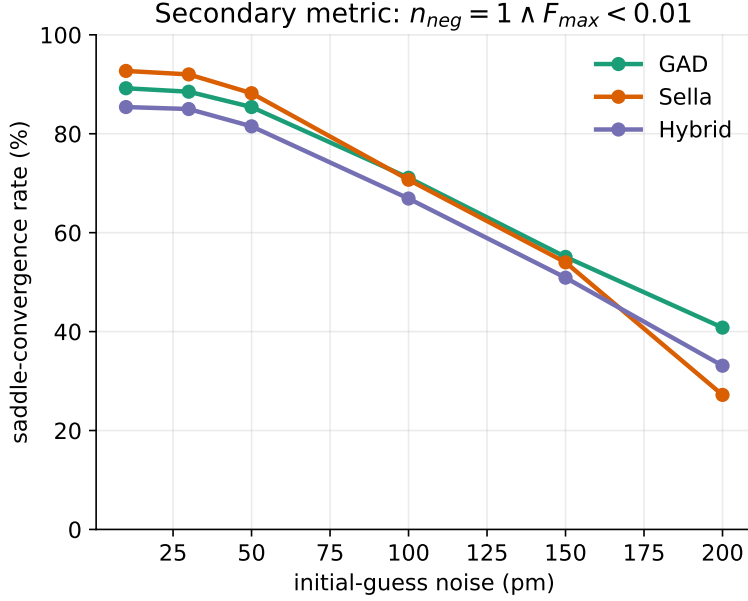


Figure 7: Secondary metric: saddle-convergence rate ($n_{\text{neg}}=1 \wedge F_{\text{max}} < 0.01$) vs. noise, $n = 287$. Sella leads at low noise; GAD overtakes at high noise — but convergence is not chemistry (cf. Fig. 1).

Table 2: Saddle-convergence rate (%), $n_{\text{neg}}=1 \wedge F_{\text{max}} < 0.01$, $n = 287$).

Saddle-convergence %	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD (ours)	89.2	88.5	85.4	71.1	55.1	40.8
Sella (baseline)	92.7	92.0	88.2	70.7	54.0	27.2
Hybrid (ours)	85.4	85.0	81.5	66.9	50.9	33.1

The F_{max} plateau.

- **Plateau.** All GAD dt collapse to 0% at $F_{\text{max}} < 0.005$ at every noise; Sella’s Newton step retains 24–34% (low noise). Pure first-order dynamics has no samples in 0.001–0.005 eV/Å (Figs. 8, 9).
- **Mechanism.** GAD step $\sim dt \|\mathbf{F}\|$ shrinks linearly as $\mathbf{F} \rightarrow 0$ and falls below the eigenvector noise floor; Newton $\sim \|\mathbf{H}^{-1}\mathbf{F}\|$ stays finite. Structural to fixed-step first-order dynamics; larger dt trades it for instability.
- **Intrinsic.** $5\times$ budget (10k vs. 2k steps, 50 pm) leaves $\tau=0.01$ unchanged (85.7%) and $\tau < 0.005$ exactly 0.0%.
- **It lives on the secondary axis.** Intended success (Section 3.1) needs no sub-plateau force, so the plateau does not affect the primary metric.

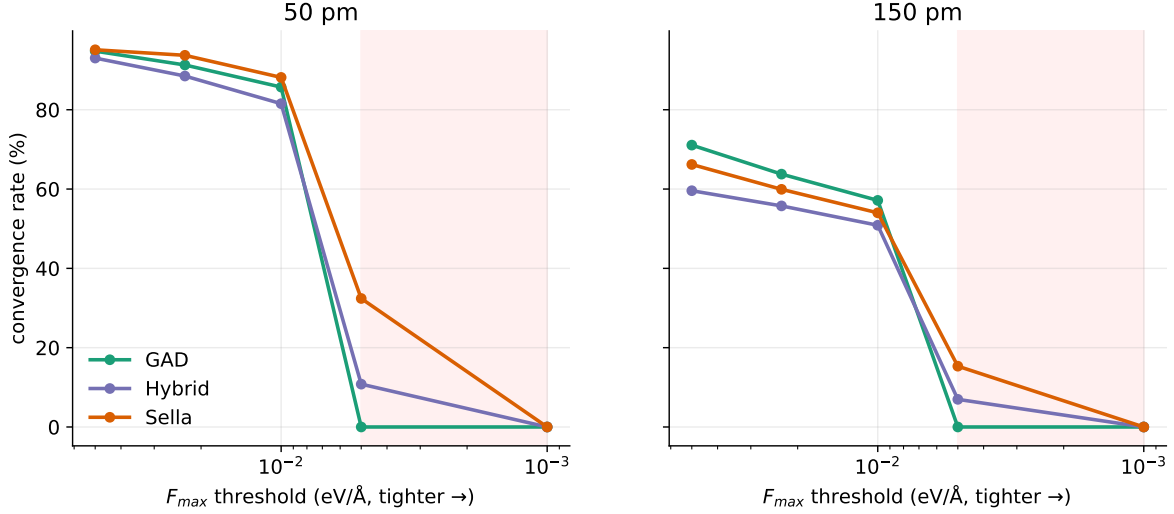


Figure 8: The $F_{\max}$ plateau (50 and 150 pm noise). Saddle-convergence rate (%), y vs. the $F_{\max}$ threshold (x , log axis, tightening to the right; shaded = the sub-0.005 plateau zone), one line per method. GAD drops vertically near $F_{\max}=0.01$ — it has essentially no converged samples below it — whereas Sella and the hybrid keep descending into the 0.005 band because their trust-radius-capped Newton step continues to drive the residual force down. No method reaches 0.001 within the 2,000-step budget.

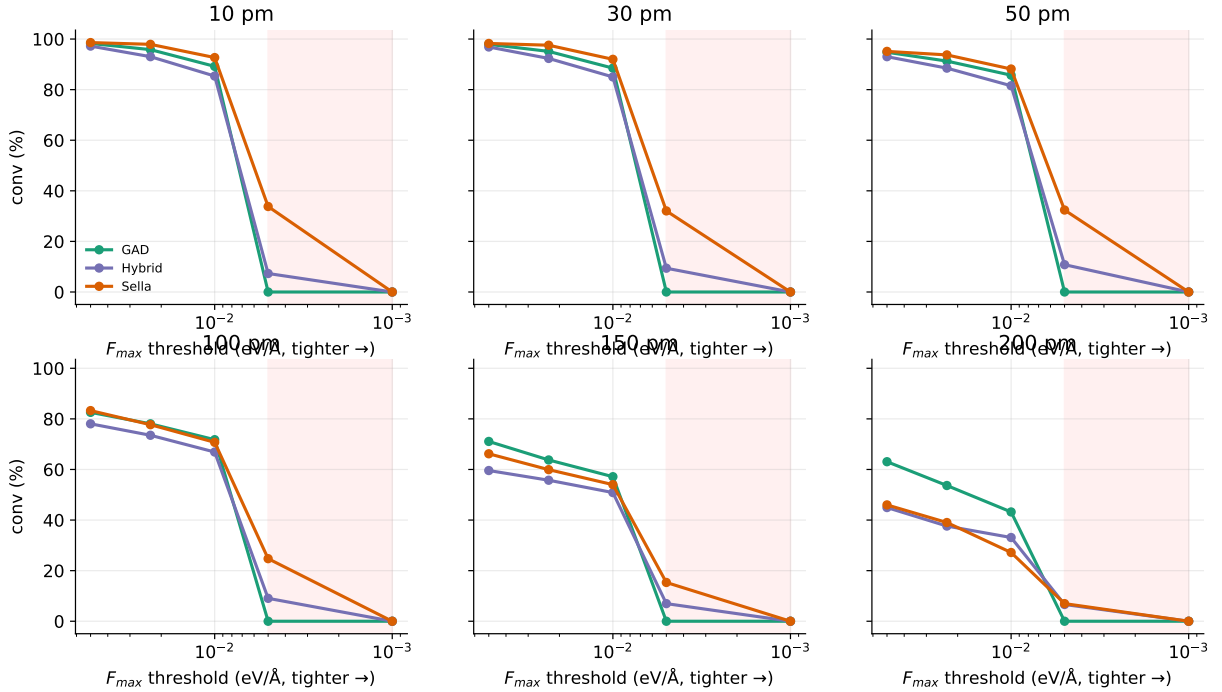


Figure 9: Saddle-convergence vs. $F_{\max}$ threshold, one panel per noise level (10–200 pm). Each panel plots convergence rate (%) against the force threshold (log x , tightening to the right; shaded = plateau zone) for GAD, the hybrid, and the Sella baseline. The GAD curve collapses vertically to 0% at $F_{\max} < 0.005$ in *every* panel, whereas Sella and the hybrid descend further — the plateau is a property of the fixed-step first-order dynamics, universal across noise.

3.4 Cost and geometry: the hybrid wins both

- **Wall-time / converged TS:** hybrid lowest at every noise — $1.3\text{--}3.0\times$ vs. Sella, $3.6\text{--}4.5\times$ vs. GAD (Fig. 10, Table 3). Few-but-expensive Sella steps (4–13) vs. many-cheap GAD steps (100–738) vs. intermediate-cheap hybrid (6–95) (Fig. 10). Pareto-dominant at high noise (Fig. 11).
- **Geometry:** converged-TS RMSD-to-reference — hybrid holds the tightest p95 at high noise (200 pm: **0.109** vs. GAD 0.456 vs. Sella 0.838 Å; $7.7\times$ tighter than Sella). Sella’s tail is bimodal (Fig. 11, Table 3).

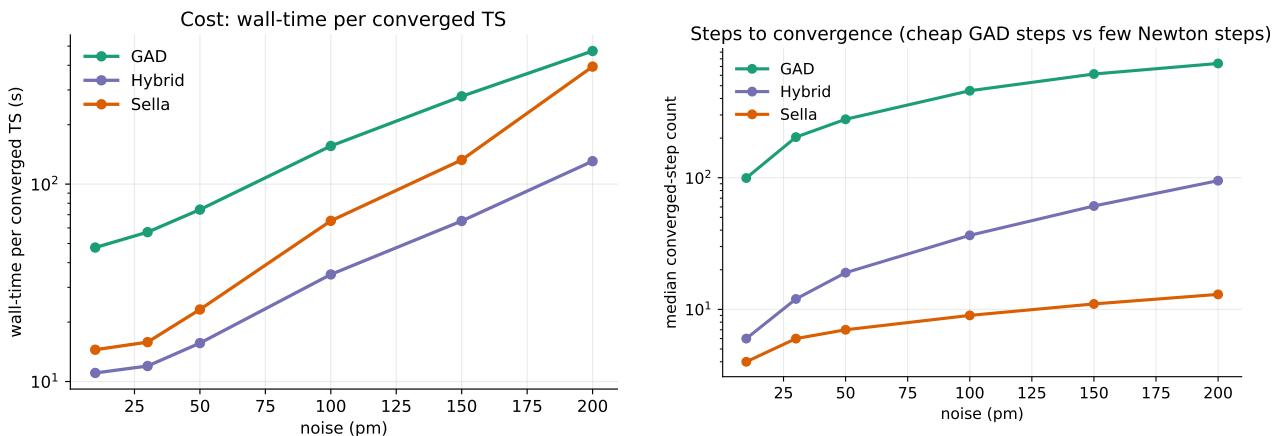


Figure 10: Cost. Wall-time per converged TS (left) and median steps (right). The hybrid is fastest despite more steps than Sella, because its steps are cheap.

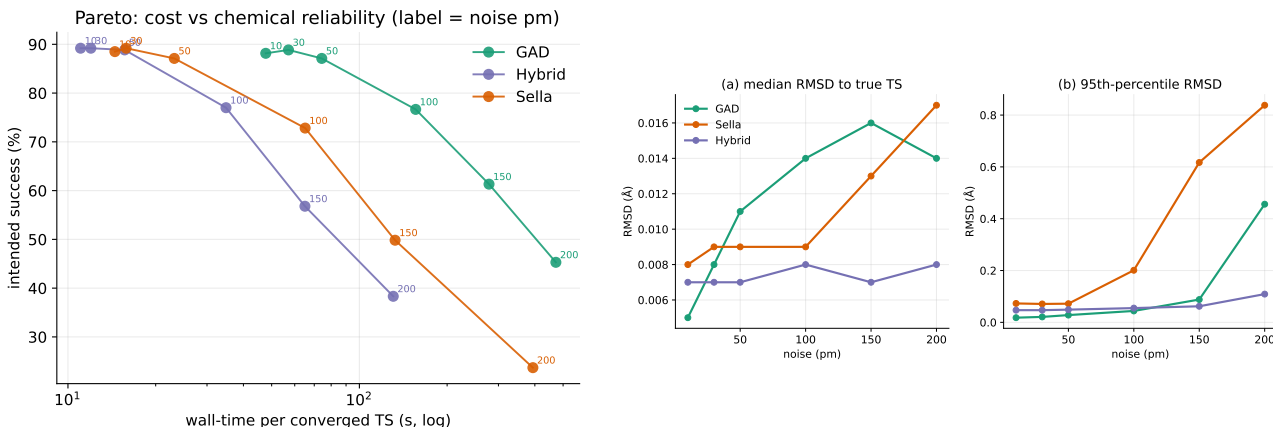


Figure 11: Left: Pareto (intended vs. wall-time; up-left is better) — the hybrid is dominant at high noise. **Right: RMSD to true TS** (median and p95) — the hybrid holds the tightest tail.

Table 3: Wall-time per converged TS (s) and converged-TS RMSD ($\text{\AA}$, p95), $F_{\max} < 0.01$, $n = 287$.

		10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
wall/conv (s)	Hybrid	11.0	12.0	15.7	34.9	65.0	130.7
	Sella	14.5	15.8	23.2	65.2	132.6	393.8
	GAD	47.7	57.1	74.3	156.0	278.5	472.3
p95 RMSD ($\text{\AA}$)	Hybrid	0.047	0.047	0.049	0.055	0.062	0.109
	Sella	0.073	0.071	0.072	0.201	0.617	0.838
	GAD	0.018	0.021	0.028	0.044	0.088	0.456

3.5 Scope boundary: GAD needs to start near the saddle

- Robustness above is for *perturbed-saddle* guesses. From a *reactant* start the GAD family is weak: hybrid $\sim 2\%$, GAD $dt=0.005$ 54% (partial n), while Sella reaches 81% (Fig. 12).
- Interpretation: far from the saddle the reaction mode is not yet identifiable as the lowest Eckart-projected eigenvector, so the GAD reflection has no correct direction to climb. GAD/hybrid are TS *refiners* given a near-saddle guess, not single-ended searchers from a minimum.

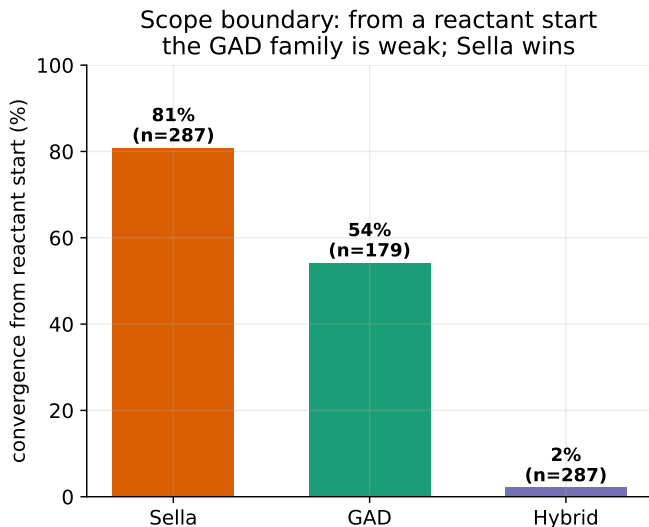


Figure 12: Scope boundary. Convergence from a reactant start: Sella wins; the GAD family needs a near-saddle guess.

4 Discussion

- When the analytic Hessian is cheap, the limiting factor is the step, not the curvature budget. A minimal first-order GAD walk is more robust on the metric that matters (intended success) precisely where large Newton/RFO jumps fail — under a degraded guess — while tying at low noise.
- The failure is *not* wrong basins: across all methods, converged geometries essentially never connect the wrong reaction. The high-noise gap is non-convergence plus one-sided (partial) saddles, which the small in-basin Euler step avoids.

- **Recommended default:** Hybrid GAD–Newton — fastest per converged TS, geometrically tightest, and chemically as reliable as GAD — for near-saddle (refinement) guesses.
- **Limitations:** (i) demonstrated for perturbed-saddle guesses; from a reactant start the GAD family is weak and Sella wins (Section 3.5). (ii) Convergence below $\tau=0.001$ is unreachable for all methods here. (iii) Isotropic perturbations; template/interpolation guesses carry structured error (future work).

Appendix

A Full tables (all nine configurations, $F_{\max} < 0.01$, $n = 287$)

Table 4: Intended-reaction success % (top) and saddle-convergence % (bottom), all nine configurations, from the aggregate IRC sweep. Bold = column best. The three headline methods (GAD, Sella cart+Eckart untuned d=1, Hybrid damped) were independently re-validated per-sample (FINDING_IRC_5TIER_2026_06_23; irc_outcomes_5tier_test287.csv); main-text numbers use that fresh re-validation and differ here by ≤ 2 pp, with unintended ≈ 0 confirmed.

Intended %	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD $dt = 0.003$	88.5	89.2	88.9	78.4	61.0	44.6
GAD $dt = 0.005$	88.9	89.2	88.9	78.0	61.7	44.6
GAD $dt = 0.007$	88.5	88.5	88.5	78.4	61.7	43.9
Sella cart tuned d=1	88.9	88.9	87.5	70.7	46.7	17.8
Sella cart+Eck untuned d=1	89.2	89.2	87.5	72.5	49.8	23.3
Sella internal tuned d=1	88.2	87.5	83.3	64.5	33.1	16.0
Sella cart+Eck untuned d=3	88.5	88.9	87.1	70.4	45.3	22.0
Hybrid damped Eckart tr=0.05	89.2	88.9	88.9	76.7	57.5	38.7
Hybrid undamped Eckart tr=0.05	88.9	88.9	88.5	77.0	57.1	38.7
Saddle-convergence %						
GAD $dt = 0.003$	89.2	88.5	85.4	71.1	55.1	40.8
GAD $dt = 0.005$	89.2	88.5	85.7	71.8	57.1	43.2
GAD $dt = 0.007$	89.2	88.9	85.7	72.8	58.2	44.6
Sella cart tuned d=1	92.0	91.3	87.5	65.5	42.9	18.8
Sella cart+Eck untuned d=1	92.7	92.0	88.2	70.7	54.0	27.2
Sella internal tuned d=1	79.1	77.4	71.8	50.9	26.8	13.9
Sella cart+Eck untuned d=3	96.5	95.5	92.0	72.8	50.5	23.3
Hybrid damped Eckart tr=0.05	85.4	85.0	81.5	66.9	50.9	33.1
Hybrid undamped Eckart tr=0.05	84.7	84.3	81.5	65.5	49.8	31.0

B Comprehensive $F_{\max}$ sweep (convergence %, $n = 287$)

Table 5: Convergence at five thresholds; GAD block is 0.0% at $F_{\max} < 0.005$ everywhere.

thr	method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
0.05	GAD $dt=0.005$	98.3	97.9	94.8	82.6	71.1	63.1
	Sella $d=1$	98.6	98.3	95.1	83.3	66.2	46.0
	Hybrid damped	97.2	96.9	93.0	78.0	59.6	44.9
0.01	GAD $dt=0.005$	89.2	88.5	85.7	71.8	57.1	43.2
	Sella $d=1$	92.7	92.0	88.2	70.7	54.0	27.2
	Hybrid damped	85.4	85.0	81.5	66.9	50.9	33.1
0.005	GAD $dt=0.005$	0.0	0.0	0.0	0.0	0.0	0.0
	Sella $d=1$	33.8	32.1	32.4	24.7	15.3	7.0
	Hybrid damped	7.3	9.4	10.8	9.1	7.0	6.6

C Convergence-criterion family

Force thresholds (eV/Å): ASE default 0.05; Gaussian “Normal” max-force 0.0231 ($4.50 \times 10^{-4} E_h/a_0$); project canonical 0.01; tight 0.005; Sella documentation 0.001. $F_{\max}$ = maximum force component on any atom.

D Sella baseline configurations

Three axes: {cartesian|internal} {Eckart}? {tuned|untuned} Hess.Freq.= N . Headline baseline: cartesian + Eckart, untuned, Hess.Freq.=1 (exact Hessian every step). Hess.Freq.=3 (library default) raises convergence at low noise (96.5 vs. 92.7% @ 10 pm) but lowers intended success at every noise (Fig. 5) — a stale Hessian yields more one-sided/failed outcomes, not unintended ones.

E Reproducibility

- **Data (committed CSVs):** master_2026_05_16.csv, noise_sweep_with_irc.csv, irc_topo_existing_methods.csv, irc_topo_hybrid.csv, threshold_sweep_2026_05_16.csv, rmsd_to_known_ts_compare.csv, reactant_0pm_2026_05_16.csv, test_summary_full.csv (per-sample).
- **Figures:** make_figs2.py. **Code:** core/gad.py (Eq. 1), projection/projection.py, search/hybrid_gad_eigfollownewton_eckart.py, search/irc_validate.py (Sella IRC + VF2).
- Headline numbers are full $n = 287$. The 200 pm count illustration (130/89/68; 68/101/116) is from the latest local re-scoring and may differ by a few counts from the committed aggregate CSVs; McNemar significance is from the per-sample paired analysis.

References

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F Figure gallery (one joint view per result)

One curated figure per result, favouring the joint multi-panel / multi-line views. Sources: `figures_new/` (fresh test-287 5-tier IRC, built by `scripts/build_figures_new.py`) and `figures/fig_master_4axis` (all-method comparison).

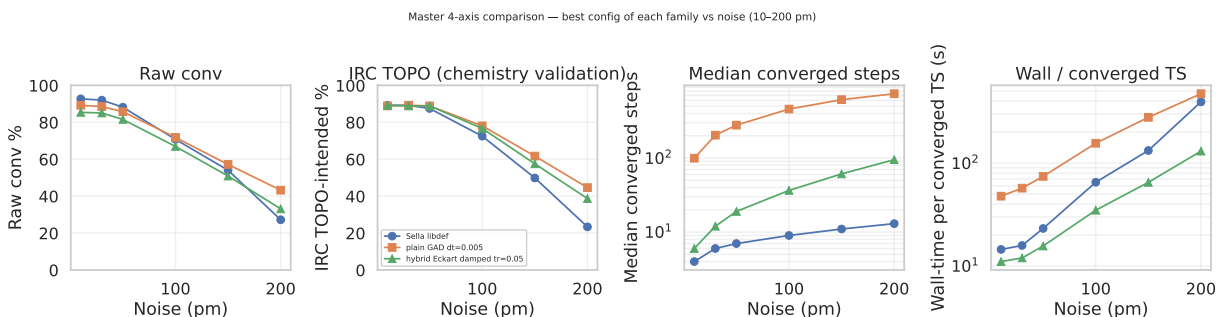


Figure 13: All-method four-axis summary (joint). Intended-reaction success, saddle-convergence, wall-time per converged TS, and RMSD-to-true-TS, all nine configurations plotted against initial-guess noise on one set of axes ($n = 287$). The single most complete view of the benchmark.

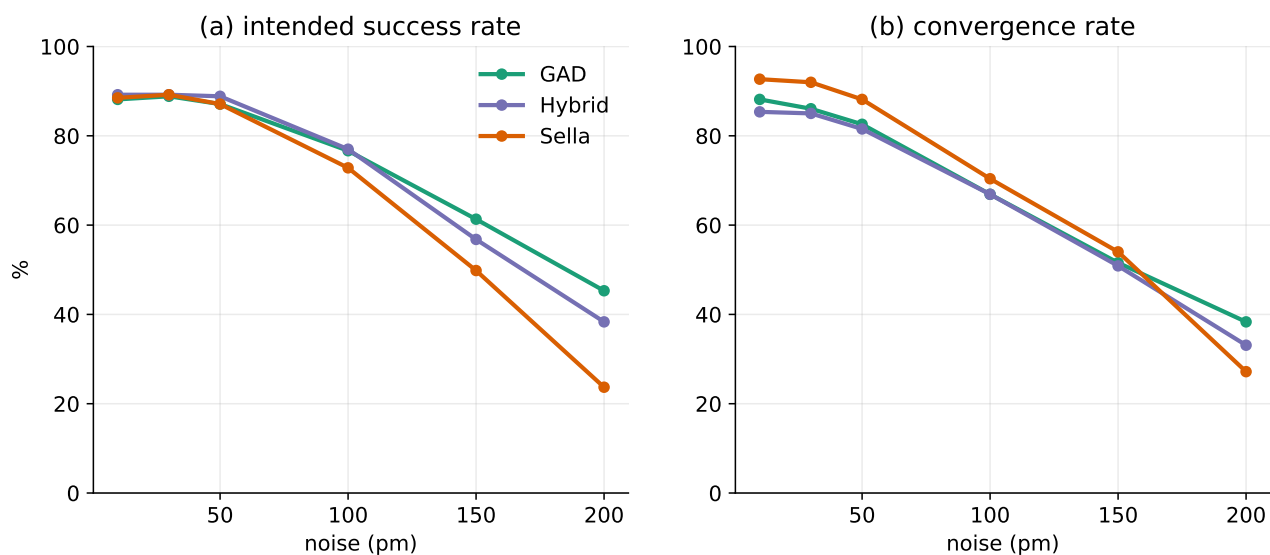


Figure 14: Primary result (focused). Intended-reaction success (left) and saddle-convergence (right) vs. noise for GAD, the hybrid, and the strongest Sella baseline. The methods tie when the guess is good and GAD pulls ahead as it degrades; the crossover holds on both axes.

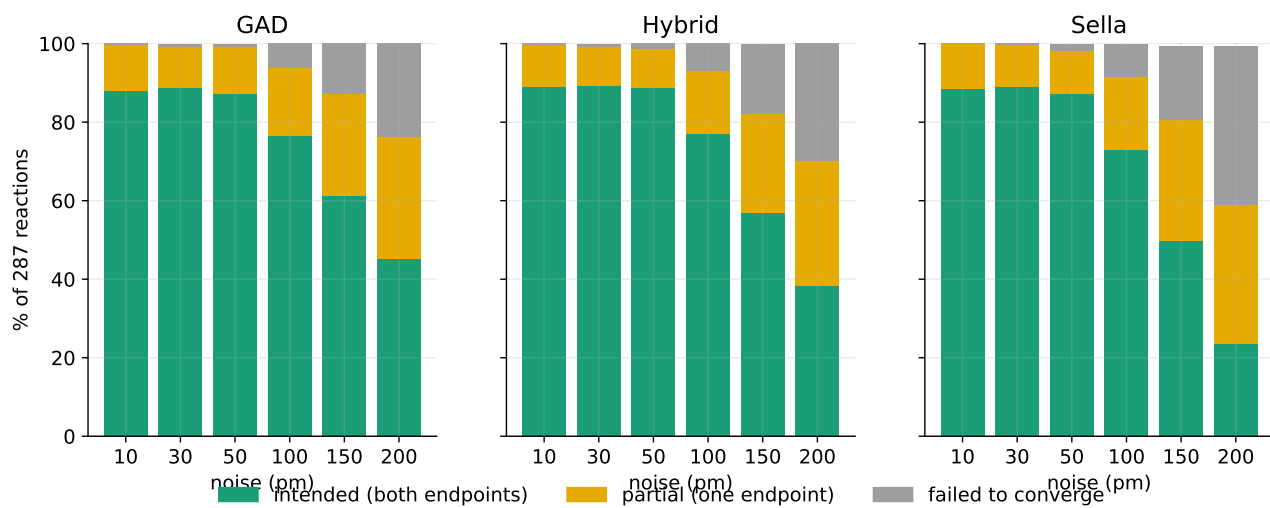


Figure 15: IRC outcome composition. Each bar splits the 287 reactions into intended, partial (one-sided), and failed-to-converge. No method produces *unintended* saddles; the high-noise gap is one-sided recovery plus non-convergence, not wrong basins.

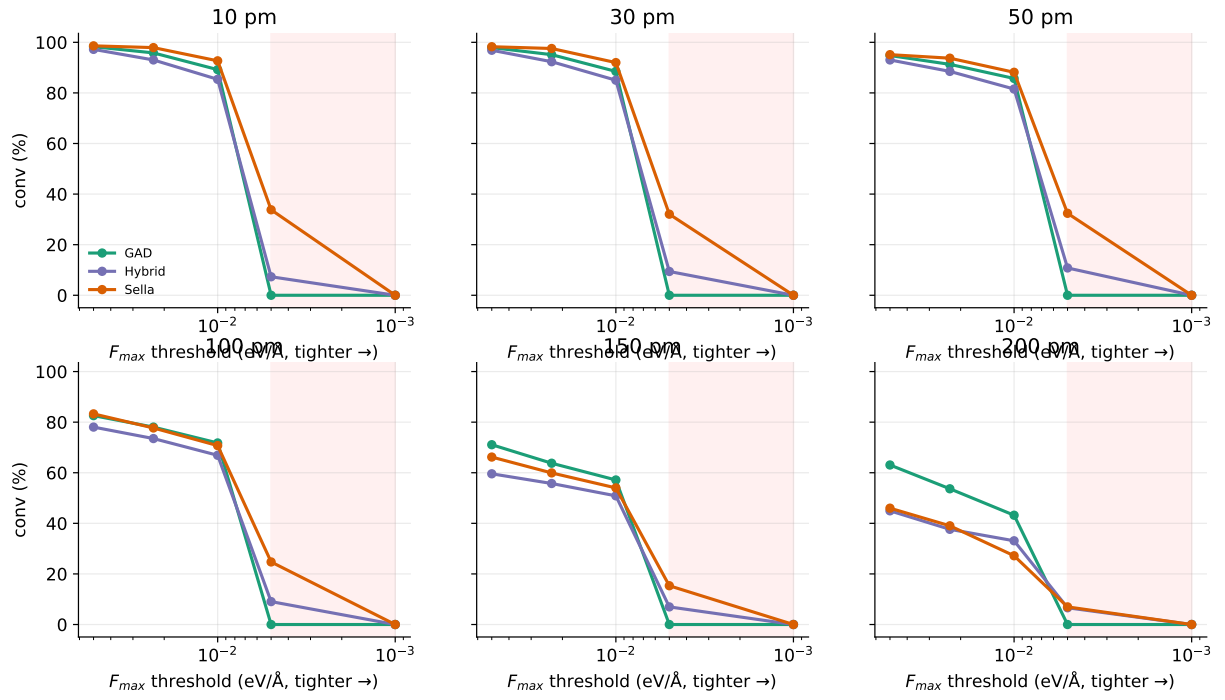


Figure 16: The $F_{\max}$ plateau across all noise levels. Convergence vs. force threshold; GAD cliffs to zero below ≈ 0.01 eV/Å at every noise level while the Newton-based methods descend further.

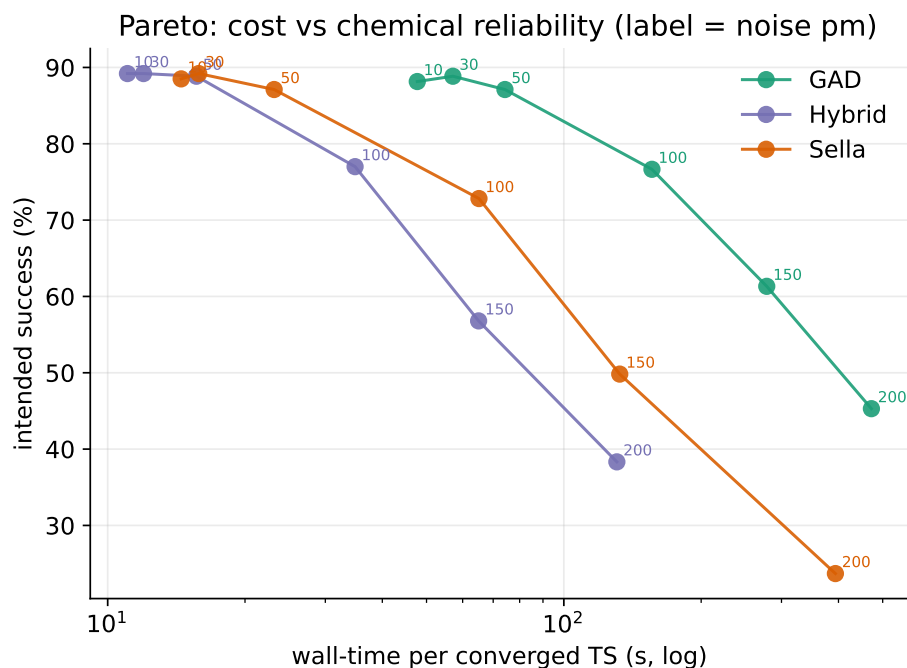


Figure 17: Cost vs. chemical reliability (joint). Wall-time per converged TS against intended success, one point per noise level per method. The hybrid is Pareto-dominant at high noise.

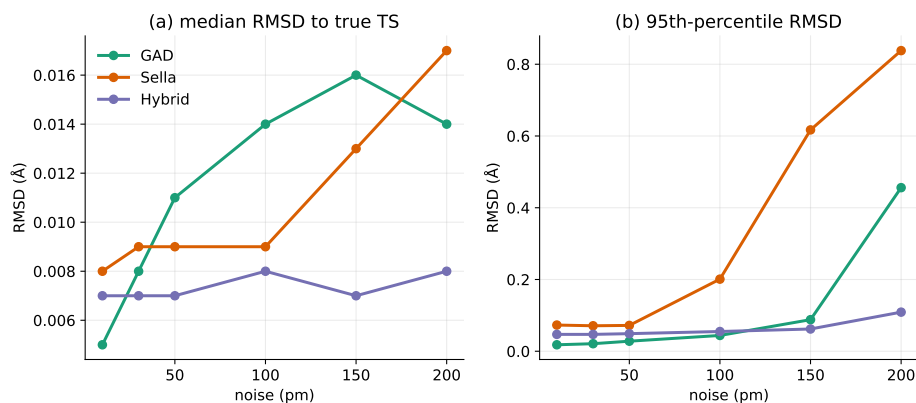


Figure 18: Geometric quality. Median and 95th-percentile RMSD of the converged TS to the labelled saddle vs. noise; the hybrid stays tight where Sella develops a large-RMSD tail.

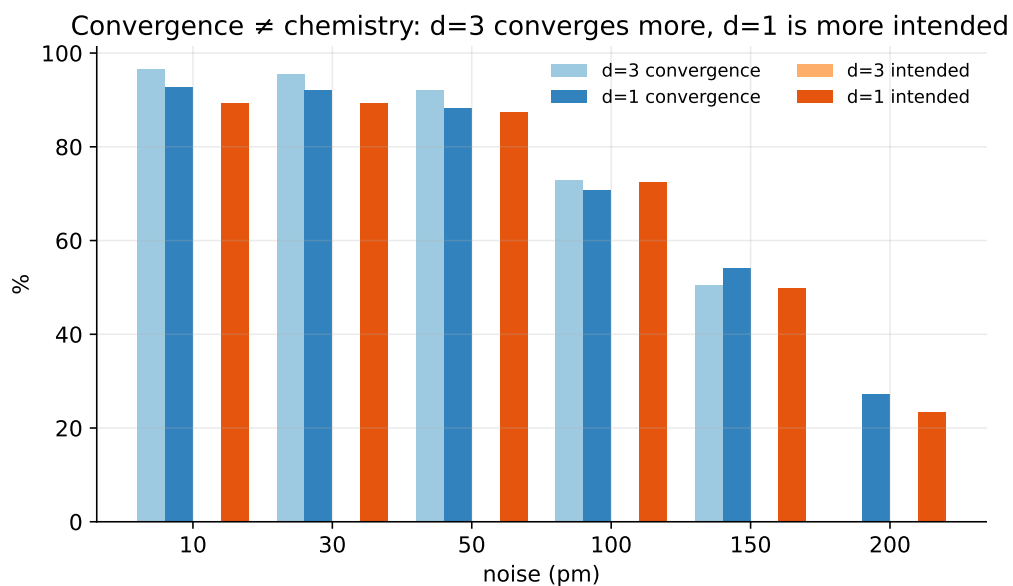


Figure 19: Convergence $\neq$ chemistry. Sella d=3 converges more often than d=1 at low noise yet recovers the intended reaction *less* often at every noise level — a stale Hessian buys convergences that are not the intended TS.

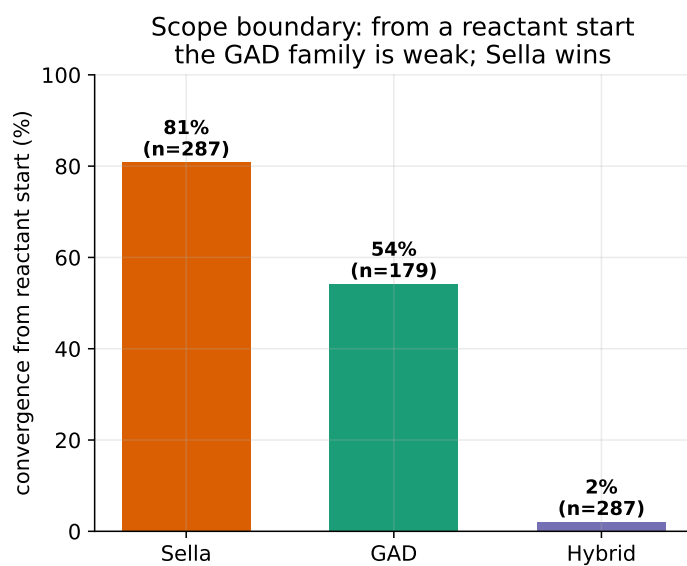


Figure 20: Scope boundary. From a reactant start the GAD family is weak and Sella wins; GAD needs a near-saddle guess for the reaction mode to be identifiable.